

RL-ILT: Pixel-Based Reinforcement Learning for Inverse Lithography Technology

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Abstract—Inverse Lithography Technology (ILT) is a critical resolution enhancement technique for advanced integrated circuit manufacturing. However, traditional ILT suffers from high computational costs, while emerging supervised deep learning approaches are severely limited by the scarcity of high-quality paired training data. Although recent reinforcement learning (RL) methods offer a data-free alternative, they typically rely on edge-based segments, limiting their ability to optimize pixel-level details and complex mask topologies. To address these challenges, we propose RL-ILT, a pixel-based deep reinforcement learning framework for mask optimization. Formulating mask synthesis as a sequential decision-making process, our agent interacts directly with a lithography simulator to optimize lithography masks. We employ a Double Deep Q-Network (Double DQN) combined with a novel action masking mechanism to restrict the search space to lithographically sensitive regions. Furthermore, we introduce a data-augmented fine-tuning strategy to adapt the pre-trained policy to unseen layouts. Experimental results demonstrate that RL-ILT outperforms state-of-the-art methods, reducing the squared L_2 error by 6% compared to the previous best approach, while generating high-fidelity masks without the need for ground-truth labels.

I. INTRODUCTION

As integrated circuit (IC) technology advances, higher demands are placed on the quality of lithography imaging in chip manufacturing. Traditional resolution enhancement techniques (RETs) can no longer meet the needs of advanced processes [1]. To address this, computational lithography has been developed, combining algorithms and physical models to improve the lithographic accuracy [2], among which optical proximity correction (OPC) aims to compensate for the printing errors caused by lithography proximity effects. Inverse Lithography Technology (ILT), as a rigorous OPC method, calculates the optimal mask pattern by solving an inverse problem based on the target design. While ILT theoretically delivers superior printability, its practical deployment is severely hindered by the prohibitive computational costs associated with iterative optimization algorithms.

With the rapid development of artificial intelligence, researchers have combined ILT with deep learning to overcome these computational bottlenecks. Supervised learning approaches have shown remarkable promise. Jiang et al. [3] proposed a fast machine learning-based mask printability prediction framework to accelerate the OPC process, significantly reducing computational overhead while ensuring comparable printability and accuracy in lithography-related applications.

Xu et al. [4] introduced an end-to-end self-training mask optimization framework based on a semantic segmentation network, which enhances mask printability and reduces training time significantly, outperforming traditional ILT and other advanced methods in both accuracy and efficiency. Yang et al. [5] developed GAN-OPC, a lithography-guided generative adversarial network framework designed for mask optimization, which leverages ILT-guided pretraining and integrates U-Net and SPSR techniques to enhance model convergence and mask quality. Zhu et al. [6] proposed L2O-ILT, a deep learning-based framework that unrolls the iterative ILT optimization algorithm into a learnable neural network with high interpretability, aiming to generate high-quality initial masks for fast refinement while overcoming runtime overhead issues associated with shrinking technology nodes. Jiang et al. [7] presented Neural-ILT, an end-to-end learning-based framework for OPC. This framework integrates domain-specific knowledge into a neural network to enhance mask printability, optimize mask complexity, and accelerate the OPC process. As a result, it achieves significant turnaround time speedup and superior mask quality compared to both traditional and other learning-based methods.

However, the success of these methods heavily relies on large-scale, high-quality paired datasets (target layouts and optimal masks). Generating such "ground-truth" labels via traditional ILT is extremely time-consuming and expensive, creating a significant data bottleneck.

To mitigate the reliance on paired data and enable simulation-driven optimization, recent studies have begun to explore reinforcement learning (RL) and hybrid approaches. Wang et al. [8] presented A2-ILT, a GPU-accelerated ILT framework, which dynamically adjusts optimization strength via a spatial attention mechanism and combines on-the-fly rectilinearization to reduce mask fragmentation. However, its core optimization still relies on differentiable gradient descent methods (e.g., L_2 error minimization), with reinforcement learning used to indirectly tune attention map hyperparameters (not direct mask optimization). This limits A2-ILT's ability to handle non-differentiable/discrete manufacturing targets (e.g., edge placement error) and its generalization under practical lithography constraints. Liang et al. [9] introduced RL-OPC, which is the first to apply deep reinforcement learning to mask optimization, enabling direct optimization of non-differentiable objectives (e.g., the process variability band)

and adaptive responses to complex process variations via end-to-end training. Yet, it adopts an edge-based modeling approach—only correcting key edge segments while ignoring pixel-level global mask structure. Though this boosts computational efficiency, it hinders fine sub-resolution assist feature (SRAF) generation and dense pattern handling, compromising mask print quality, especially in extreme ultraviolet (EUV) or high numerical aperture (high-NA) lithography.

To bridge this gap and overcome the dimensionality challenge, we propose RL-ILT, a pixel-based deep reinforcement learning framework for mask optimization. By formulating mask synthesis as a sequential decision-making process, our agent interacts directly with a lithography simulator. This allows for the direct optimization of complex manufacturing objectives through a reward mechanism, bypassing the need for ground-truth labels. Crucially, we introduce a novel Action Masking mechanism to dynamically restrict the search space to lithographically sensitive regions, ensuring both convergence efficiency and mask manufacturability. Furthermore, a data-augmented fine-tuning strategy is employed to enable rapid adaptation to unseen layouts.

The primary contribution of this work is the establishment of a pixel-based reinforcement learning paradigm for ILT that breaks the dependency on ground-truth data while maintaining high mask fidelity. Specifically, our main contributions are summarized as follows:

- We utilize reinforcement learning to build a pixel-based ILT framework RL-ILT for mask optimization and apply a multi-objective double deep Q-network (Double DQN) to optimize pattern quality.
- A two-stage optimization framework comprising large-scale pre-training followed by data-augmented test-time fine-tuning is applied to RL-ILT, which facilitates rapid adaptation to previously unseen layout configurations.
- The empirical results demonstrate that RL-ILT achieves superior performance compared to the current state-of-the-art method, as measured by metric squared L_2 error on industrial designs provided by ICCAD 2013 CAD Contest.

The rest of the paper is organized as follows. Section II introduces the preliminaries. Section III discusses our algorithms and framework in detail. Section IV presents experimental results, followed by a conclusion in Section V.

To enhance the reproducibility, replicability, and transparency of this work, the complete source code and relevant implementation details have been publicly released on GitHub: <https://github.com/Nemophilist-14514/RL-ILT>.

II. PRELIMINARIES

In this section, we present the conceptual foundation of Inverse Lithography Technology and formally define the problem addressed in this paper.

A. Inverse Lithography Technology

ILT aims to find an optimal mask pattern ψ^* such that the resulting wafer image ω closely matches a given target layout ϕ . This process is governed by a forward lithographic operator

f , which models the complex physical transformation from the mask to the wafer, i.e., $\omega = f(\psi)$. The operator f encapsulates various physical phenomena, including the three-dimensional electromagnetic interactions of the mask, the optical properties of the illumination source and projection lens, the chemical behavior of the photoresist, as well as process variations such as dose and focus fluctuations [10], [11].

Formally, the ILT problem can be viewed as seeking the inverse of this operator: $\psi^* = f^{-1}(\phi)$. However, this inverse problem is fundamentally ill-posed for two primary reasons. First, the forward operator f is many-to-one, which means numerous distinct mask patterns can produce nearly identical wafer images due to the low-pass filtering nature of the optical system. Second, idealized target layouts ϕ —often composed of sharp corners and Manhattan geometries—typically cannot be perfectly replicated on the wafer under real-world lithographic constraints.

To address these issues, the ILT problem is reformulated as an optimization task. A cost function $H(\psi, \phi)$ is defined to quantify the quality of a candidate mask ψ . A simple example of such a function is the squared L_2 norm between the simulated wafer image and the target:

$$H(\psi, \phi) = \iint \|f(\psi) - \phi\|_2^2 dx dy. \quad (1)$$

In practice, $H(\psi, \phi)$ is often more sophisticated. It often incorporates additional terms to ensure robustness across the process window (e.g., multiple dose and focus conditions), to enhance image contrast (e.g., using Normalized Image Log-Slope, NILS), and to account for other critical factors like the Mask Error Enhancement Factor (MEEF) [12], [13].

Crucially, the cost function may also include non-lithographic objectives, such as a preference for simpler, more manufacturable masks over highly complex ones. The ultimate goal is to find a mask ψ^* that minimizes $H(\psi)$ while adhering to practical mask manufacturing constraints [14].

B. Problem Formulation

We formulate the mask optimization problem as follows. Given a target layout ϕ and a lithography process model f , the mask optimization problem is to find an optimized mask ψ that minimizes the squared L_2 error against the target.

Despite its theoretical soundness, practical ILT deployment faces several critical challenges:

- **Ill-posedness and Non-Uniqueness:** As mentioned, the inverse problem has infinitely many solutions, making it difficult to converge to a single, optimal mask without strong regularization.
- **Data Scarcity for Learning-Based Methods:** Supervised deep learning approaches require large datasets of paired inputs (target layouts) and outputs (optimal masks). Generating these "ground-truth" masks via traditional ILT is itself computationally intensive, and validating them through wafer experiments is even more costly, leading to a severe scarcity of high-quality training data.

These challenges underscore the need for novel optimization frameworks capable of efficiently navigating the high-dimensional design space without relying on scarce paired

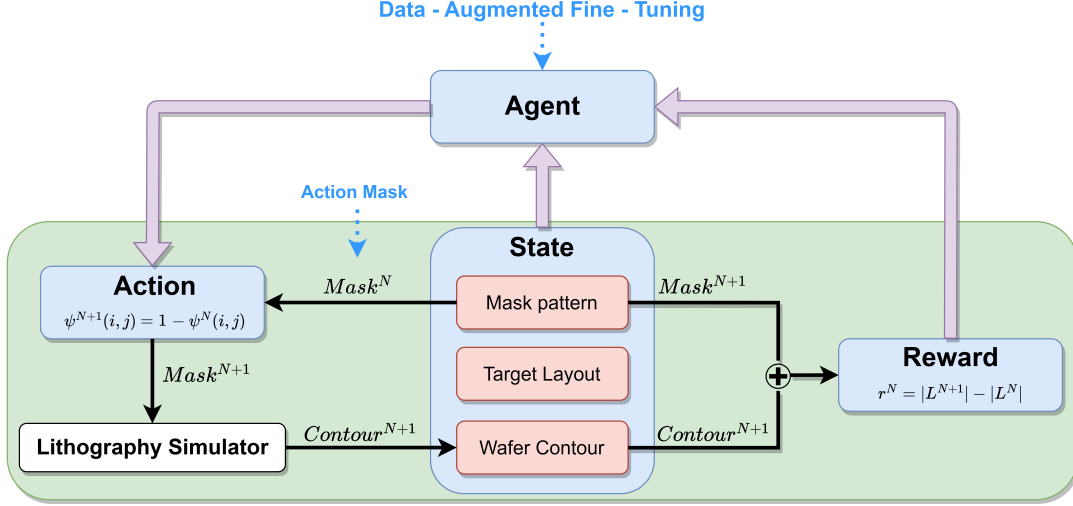


Fig. 1. Conception of RL-ILT

data. Our proposed RL-ILT framework is specifically designed to overcome these limitations.

III. RL-ILT FRAMEWORK AND ALGORITHM

Building on these foundations, we now present our RL-ILT framework. In this section, we will detail how we incorporate the RL architecture to solve ILT problem.

A. RL-ILT Framework

In RL-ILT, the RL agent is trained to sequentially flip the light transmittance of the mask pattern in order to optimize the mask. The key concepts of RL-ILT formulation in our implementation are illustrated below:

- 1) State (s): A state s represents a triplet (ϕ, ψ, ω) , which includes the target layout ϕ , the mask pattern ψ and the wafer contour ω generated by the current ψ and the provided lithography function f .
- 2) Action (a): An action a refers to a coordinate (x, y) , which represents the center of pixel seal to flip light transmittance in mask pattern. Because frequent single-pixel flips may result in overly complex masks, we introduce a local operation of flipping 3×3 blocks in our action design.
- 3) Reward (r): Reward r corresponds to the mask quality improvement after applying action a at state s . Our reward function is designed as follows:

$$r_t = |L_{t-1}| - |L_t| \quad (2)$$

where L_t means squared L_2 error between the target layout ϕ and the wafer contour ω at timestep t .

Our RL-ILT framework is shown in Figure 1 and the training procedure is presented in algorithm 1. We utilize U-net [15] to calculate the approximation of the Q-function, leveraging its symmetric encoder-decoder architecture with skip connections to capture multi-scale features and adopt Double DQN [16]

Algorithm 1 Training process of RL-ILT

Input: f : lithography function; ϕ^i : target patterns; N : number of training targets; T : maximum number of steps; γ : discount factor.

Output: θ : policy network parameters.

- 1: Replay buffer $B \leftarrow \{\}$;
- 2: Initialize policy network parameters θ randomly;
- 3: **for** each $i \in [0, N]$ **do**
- 4: Initialize mask $\psi_0^i \leftarrow \phi^i$;
- 5: Initialize wafer contour $\omega_0^i \leftarrow f(\psi_0^i)$;
- 6: Initialize state $s_0^i = (\phi^i, \psi_0^i, \omega_0^i)$;
- 7: **for** each $t \in [1, T]$ **do**
- 8: Compute action mask m_t^i according s_t^i ;
- 9: Pick action $a_t^i \leftarrow \argmax_a Q(s_t^i, a; \theta) \odot m_t^i$;
- 10: Modify mask ψ_{t+1}^i by a_t^i ;
- 11: Generate wafer contour $\omega_{t+1}^i \leftarrow f(\psi_{t+1}^i)$;
- 12: Encode state $s_{t+1}^i = (\phi^i, \psi_{t+1}^i, \omega_{t+1}^i)$;
- 13: Calculate reward $r_t^i = |L_{t-1}^i| - |L_t^i|$;
- 14: Put transition $(s_t^i, a_t^i, r_t^i, s_{t+1}^i)$ into replay buffer B ;
- 15: Sample batches from replay buffer B and training network parameters θ ;
- 16: **end for**
- 17: **end for**
- 18: **return** θ ;

as the main training strategy, which effectively mitigates the overestimation bias of Q-values by decoupling the action selection from the value evaluation.

RL-ILT takes the target layout as input. Then the agent initializes the mask pattern and flip the transmittance of the mask iteratively and sequentially. Our framework, respectively, performs the following steps in each iteration:

- 1) Initialization environment: Initialize the mask pattern ψ_0 to be consistent with the target layout ϕ , generate the wafer contour ω_0 and encode the initial state

$$s_0 = (\phi, \psi_0, \omega_0).$$

- 2) Decision action: Decide the next action a_t and output the next mask pattern ψ_{t+1} according to s_t and a_t at timestep t . Then the mask pattern ψ_{t+1} is input into the lithography function f to generate the current wafer contour ω_{t+1} . At this moment the next state is generated with the format $s_{t+1} = (\phi, \psi_{t+1}, \omega_{t+1})$.
- 3) Calculation reward: Calculate the corresponding reward r_t and put the transitions (s_t, a_t, r_t, s_{t+1}) into the replay buffer of the RL-ILT framework.
- 4) Training model: Randomly sample $(s_{t'}, a_{t'}, r_{t'}, s_{t'+1})$ from replay buffer and update the network parameters, repeat steps 2-4 until the end of training.

B. Action Masking Mechanism

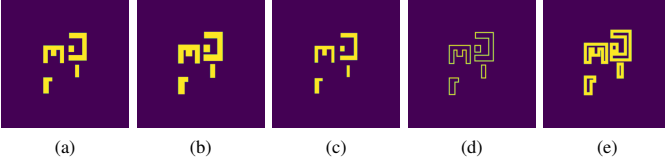


Fig. 2. Process of generating action mask: (a) Target, (b) Dilating, (c) Eroding, (d) Edge, (e) Wide Edge

To enhance learning efficiency and ensure physical plausibility of the generated masks, we introduce an action masking mechanism that dynamically restricts the set of allowable actions at each RL step. Specifically, given a target layout ϕ , we first compute its binary edge map E . The action mask $M \in \{0, 1\}^{H \times W}$ is then defined as a binary spatial mask where $M_{i,j} = 1$ if pixel (i, j) lies within a predefined neighborhood of any edge in E , and $M_{i,j} = 0$ otherwise. During policy execution, only pixels with $M_{i,j} = 1$ are eligible for flipping. All other actions are masked out by setting their log-probabilities to $-\infty$ before sampling.

The process of generating action mask M is shown in Fig 2. It begins by expanding (dilating) and shrinking (eroding) the input image using a small structuring element. The difference between these two transformed images yields initial edge information, capitalizing on the morphological gradient principle where boundary regions exhibit the most significant intensity variation between expanded and contracted structures. This edge map is then processed with a larger structuring element to widen edges, followed by thresholding to convert the result into a binary mask, ensuring clear, connected edge representation.

Fig 3 shows the best L_2 loss in per training episode, where the size of sliding window is 20. The training curves demonstrate that incorporating Action Mask significantly improves optimization stability and convergence performance. Compared with the baseline without Action Mask, the proposed method achieves consistently lower L_2 loss throughout training and exhibits reduced variance during later training stages, indicating more stable policy updates and more efficient exploration for generating better masks.

This design is motivated by two key observations from lithographic imaging physics. First, modifications to mask regions

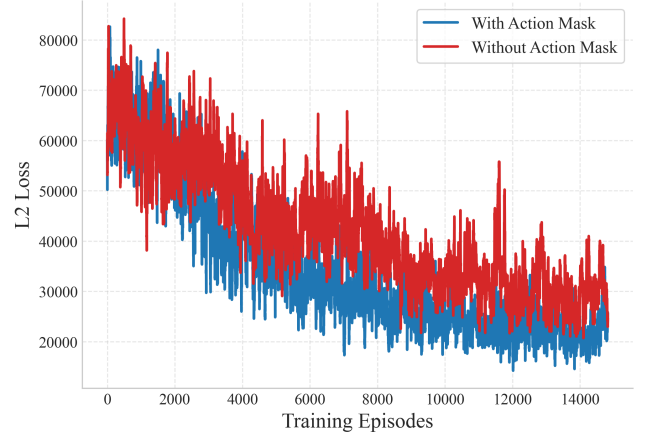


Fig. 3. Comparisons of Best Mask per Episode

far from pattern edges—such as large solid opaque or clear areas—have negligible impact on the printed image due to optical low-pass filtering [14], [17]. Second, unrestricted pixel-wise optimization over the entire mask often leads to high-frequency noise or non-manufacturable serpentine patterns that degrade robustness under process variations. By confining agent actions to edge-proximal zones, our method focuses exploration on the most sensitive and lithographically relevant regions. This not only accelerates convergence by reducing the effective action space by orders of magnitude but also inherently promotes mask smoothness and manufacturability. Empirically, as shown in Section IV, ablating the action mask leads to inferior final image fidelity, confirming its critical role in guiding effective policy learning.

C. Data-Augmented Fine-Tuning

Motivated by some Pre-training and Fine-tuning learning approaches [18]–[20] that leverage large-scale datasets to capture universal representations to enhance the robustness of specialized models, we explore the utilization of Data-Augmented Fine-Tuning in RL-ILT. While the policy network is pre-trained on a large-scale synthetic dataset covering diverse layout patterns and process conditions, its generalization capability may degrade when applied to unseen or highly specialized target layouts (e.g., dense logic cells or custom SRAM macros). To bridge this domain gap, we introduce a fine-tuning stage that leverages on-the-fly data augmentation during inference preparation.

Specifically, given a new target layout ϕ_{target} , we generate 8 augmented variants $\{\phi_{\text{aug}}^{(k)}\}_{k=1}^8$ by applying pattern rotation and reflection for symmetric structures. In each testing epoch, the target image is cloned and subjected to horizontal flipping twice. Subsequently, each flipped image undergoes rotational transformations of 90° , 180° , 270° , and 360° , resulting in the generation of 8 augmented image variants for each original target image.

For each augmented target $\phi_{\text{aug}}^{(k)}$, we roll out trajectories using the pre-trained policy and collect transition tuples (s_t, a_t, r_t, s_{t+1}) from the lithography simulator. These samples are then used to perform a few-step policy update via Double

TABLE I
PERFORMANCE COMPARISONS WITH OTHER METHODS ON SQUARED L_2 ERROR

Benchmarks ID	MOSAIC [21] L_2 (nm^2)(↓)	PGAN-OPC [5] L_2 (nm^2)(↓)	Neural-ILT [7] L_2 (nm^2)(↓)	DevelSet [22] L_2 (nm^2)(↓)	A2-ILT [8] L_2 (nm^2)(↓)	Ours L_2 (nm^2)(↓)
case1	49893	52570	49817	49142	44477	44682
case2	50369	42253	38174	34489	34031	33553
case3	81007	83663	89411	93498	88901	87937
case4	20044	19965	16744	18682	17464	12787
case5	44656	44733	45598	44256	35685	36306
case6	57375	46062	43836	41730	41432	34008
case7	37221	26438	20324	25797	21901	19552
case8	19782	17690	13337	15460	15257	12372
case9	55399	56125	49401	50834	45885	40088
case10	24381	9990	8511	10140	11121	7964
Average ¹	44012.7	39948.9	37515.3	38402.8	35615.4	32924.9
Ratio ²	1.34	1.21	1.14	1.17	1.08	1.00

¹ The mean values from case1 to case10

² The ratio of the current average value to the optimal average value

DQN effectively adapting the policy to the local geometry and process sensitivity of ϕ_{target} .

Importantly, this fine-tuning is instance-specific but computationally efficient, it actively refines the decision policy using simulated experience derived from the target itself, thereby enabling test-time adaptation without retraining from scratch. The adapted policy is then used for final mask generation, yielding masks that are both high-fidelity and layout-aware.

IV. EXPERIMENTS DESIGN AND RESULTS

The proposed RL-ILT is verified with Pytorch and CUDA on one Linux server with Intel Xeon Scale 8358 CPU and NVIDIA A100 Tensor Core GPU. The training dataset is obtained from the MetalSet of Lithobench [23] while the testing dataset is from ICCAD 2013 contest IBM benchmarks [24]. The MetalSet subset within the LithoBench benchmark is specifically designed to support deep learning-based inverse lithography technology research for metal layers in semiconductor manufacturing. It comprises 16,472 synthetically generated layout tiles, each of size $2048\text{ nm}^2 \times 2048\text{ nm}^2$, created by adhering to the design rules of the widely used ICCAD-13 benchmark—originally derived from 32nm industrial layouts. All methods operate under the same simulation conditions provided by Lithobench. We benchmark our approach against state-of-the-art methods targeting metal-layer ILT, including gradient-based ILT and learning-enhanced frameworks.

The commonly used evaluation metrics in mask optimization is printability, which characterize the quality of the printed wafer patterns. In this paper, we adopt squared L_2 error to measure printability of the mask. Squared L_2 error is calculated by $\|f(\psi) - \phi\|_2^2$ in this paper, where f represents the forward lithography process, ψ represents the mask pattern and ϕ is the target pattern as mentioned before.

A. Comparison with the State-of-the-art Works

To test the performance of RL-ILT, we compared RL-ILT with other recent works on the test set, and the experimental results are shown in the Table I. We can see that RL-ILT outperforms all other methods in image transfer fidelity.

Specifically, compared with the previous best results, the squared L_2 error is reduced by 6%.

This superior performance stems from several key innovations in our framework. First, unlike supervised learning methods that rely heavily on paired training data, our reinforcement learning framework enables direct optimization of lithographic performance through interaction with a physical simulator. This eliminates the need for costly ground-truth mask data and allows the agent to learn robust policies that maximize long-term print quality.

Second, we meticulously design our pixel-level ILT reinforcement learning environment to balance optimization power with manufacturability. Specifically, by defining the state as a triplet (ϕ, ψ, ω) and actions as localized 3×3 pixel flips, the agent can perform fine-grained, global mask adjustments, overcoming the structural limitations of edge-based approaches like RL-OPC. The reward function, based on L_2 error, provides direct feedback from the lithography physics, guiding the agent towards high-fidelity solutions.

Furthermore, the integration of action masking is crucial for efficient learning. By restricting the agent's actions to regions near layout edges, where changes have the most significant impact on imaging, the search space is dramatically reduced. This focused exploration prevents the agent from wasting computational effort on irrelevant background areas, leading to faster convergence and higher-quality masks, as confirmed by our ablation studies.

Finally, our proposed two-stage training paradigm, large-scale pre-training followed by data-augmented test-time fine-tuning, effectively bridges the gap between synthetic training data and real-world unseen industrial layouts. This strategy allows the model to rapidly adapt its policy to the specific geometric and lithographic challenges of a new target pattern, which is a key factor in achieving state-of-the-art generalization and performance on the ICCAD benchmarks.

It should be noted that our method is specifically designed for metal layer mask synthesis, where dense line-space patterns and complex 2D features dominate. In contrast, RL-OPC [9] focuses on contact/via layers, which consist primarily of sparse, isolated circular or rectangular holes. Moreover, RL-OPC adopts a mask representation based on edge movements

TABLE II
ABLATION EXPERIMENT OF DATA-AUGMENTED FINE-TUNING AND ACTION MASK ON SQUARED L_2 ERROR

Benchmarks ID	w/o F&A L_2 (nm^2)(↓)	w/ F w/o D&A L_2 (nm^2)(↓)	w/ F&D w/o A L_2 (nm^2)(↓)	w/o F w/ A L_2 (nm^2)(↓)	w/ F&A w/o D L_2 (nm^2)(↓)	w/ F&D&A L_2 (nm^2)(↓)
case1	71929	57911	51412	79589	50214	44682
case2	55976	46550	42341	59515	32447	33553
case3	117477	112296	97562	127367	106509	87937
case4	28560	22431	18982	30032	14724	12787
case5	54791	47220	43514	63434	36873	36306
case6	52038	46103	42334	67082	35221	34008
case7	41579	34883	29267	41290	18259	19552
case8	25707	17695	14761	26954	11497	12372
case9	79722	55722	44503	71272	40504	40088
case10	25100	14412	10098	24574	7644	7964
Average	55287.9	45522.3	39477.4	59110.9	35389.2	32924.9
Ratio	1.68	1.38	1.20	1.80	1.07	1.00

and rectangular primitive decomposition, which is well-suited for the simple, Manhattan-style geometries of via/contact layers but cannot easily generalize to the irregular, non-rectangular, and highly connected structures typical of metal interconnects. Due to these fundamental differences—in pattern topology, optimization objectives, lithographic challenges, and mask parameterization—a direct quantitative comparison between RL-ILT and RL-OPC is neither technically meaningful nor standard practice in the OPC/ILT literature.

B. Ablation Experiments

To validate the effectiveness of Data-Augmented Fine-Tuning and Action Mask, the experiment set up six comparative scheme.¹

Table II presents a comprehensive ablation study to disentangle the contributions of Data-Augmented Fine-Tuning and the Action Masking mechanism. We systematically evaluated six different configurations, ranging from the baseline without any enhancement to our proposed full framework.

1) *Effectiveness of Fine-Tuning*: First, comparing the baseline (w/o F&A) with the fine-tuning-only setup (w/ F w/o D&A), the average Squared L_2 error decreases significantly from 55,287.9 nm^2 to 45,522.3 nm^2 . This substantial improvement confirms that while the pre-trained policy captures general lithographic rules, there exists a domain gap when applied to unseen test layouts. Fine-tuning allows the agent to adapt its policy to the specific geometric characteristics of the target instance.

2) *Impact of Data Augmentation*: Furthermore, incorporating data augmentation into the fine-tuning process (w/ F&D w/o A) further reduces the error to 39,477.4 nm^2 , compared to fine-tuning alone (w/ F w/o D&A, 45,522.3 nm^2). By generating rotated and flipped variants of the target pattern, data augmentation prevents the agent from overfitting to a specific orientation and encourages the learning of robust, rotation-invariant features, thereby enhancing the stability of the optimization.

¹baseline model without fine-tuning and without action mask(w/o F&A); incorporating fine-tuning but no data-augmentation and action mask (w/ F w/o D&A); incorporating fine-tuning and data-augmentation but no action mask(w/ F&D w/o A); incorporating action mask but no fine-tuning(w/o F w/ A); incorporating action mask and fine-tuning but no data-augmentation(w/ F&A w/o D) and with fine-tuning, data-augmentation and action mask(w/ F&D&A).

3) *Crucial Role of Action Masking*: Most importantly, the integration of the Action Masking mechanism proves critical for achieving state-of-the-art performance. When applied to the data-augmented fine-tuning model (comparing w/ F&D w/o A with our full method w/ F&D&A), the Action Mask reduces the average error from 39,477.4 nm^2 to the best-reported value of 32,924.9 nm^2 .

It is worth noting that applying Action Mask without fine-tuning (w/o F w/ A) yields the highest error (59,110.9 nm^2). This occurs because the action mask, without fine-tuning, imposes a rigid constraint that prevents the agent from exploring compensatory solutions outside the predefined sensitive regions. While the pre-trained policy captures general lithographic rules, it inevitably suffers from a domain gap when applied to unseen layouts. Without the adaptability provided by fine-tuning, the Action Mask imposes a hard constraint that severely limits the agent’s compensatory exploration capabilities. If the pre-trained agent’s optimal policy for a novel geometry slightly deviates from the pre-defined edge neighborhood, the mask prematurely truncates the search space, trapping the agent in sub-optimal states.

However, the true potential of the Action Mask is unlocked when coupled with Fine-Tuning. In this synergy, the mask shifts its role from a rigid constraint to a strong inductive bias. By pruning the immense action space (eliminating irrelevant background pixels), it compels the fine-tuning process to concentrate gradient updates exclusively on lithographically sensitive regions. This prevents the instance-adaptation process from wasting computational budget on the background or introducing non-manufacturable noise, thereby enabling the agent to converge rapidly to a high-fidelity solution.

In summary, the combination of all three components (w/ F&D&A) achieves a $1.68 \times$ improvement over the baseline, demonstrating that the synergy between a focused action space (Action Mask) and robust instance-adaptation (Data-Augmented Fine-Tuning) is essential for high-fidelity mask synthesis.

C. Impact of Double DQN and flip block size

To verify the impact of the learning algorithm and action granularity on optimization performance, we conducted ablation studies on the Double DQN mechanism and the size of

TABLE III
COMPARISONS OF DOUBLE DQN AND FLIP BLOCK SIZE ON SQUARED L_2 ERROR

Benchmarks	Double DQN		Flip Block Size		
	w/o	w/	1×1	3×3	5×5
case1	47995	44682	90538	44682	114714
case2	33928	33553	88644	33553	73535
case3	103771	87937	148506	87937	91343
case4	11237	12787	82080	12787	21934
case5	33093	36306	78843	36306	70236
case6	32888	34008	72671	34008	59683
case7	19356	19552	59297	19552	82820
case8	11041	12372	33440	12372	40345
case9	39672	40088	84149	40088	103144
case10	7806	7964	29602	7964	32323
Average	34078.7	32924.9	76777.0	32924.9	69007.7
Ratio	1.04	1.00	2.33	1.00	2.10

the pixel flip block. The quantitative results are presented in Table III.

First, regarding the reinforcement learning algorithm, we compared the standard Deep Q-Network (DQN) with the Double DQN employed in our framework. As shown in the "Double DQN" columns of Table III, the model incorporating Double DQN consistently achieves lower Squared L_2 error across most test cases, reducing the average error from $34,078.7nm^2$ (w/o Double DQN) to $32,924.9nm^2$ (w/ Double DQN). This improvement is attributed to Double DQN's ability to decouple action selection from target Q-value evaluation, thereby effectively mitigating the overestimation bias inherent in standard Q-learning. This leads to more stable training dynamics and a policy that converges to higher-quality mask patterns.

Second, we investigated the effect of action space granularity by varying the size of the flip block among 1×1 , 3×3 , and 5×5 pixels. The results in Table III demonstrate that the 3×3 block size yields the optimal performance, achieving the lowest average L_2 error of $32,924.9nm^2$. In contrast, the 1×1 setting results in the worst performance (average error $76,777.0nm^2$). This suggests that single-pixel updates are too granular, leading to an excessively large search space and inefficient exploration. The agent struggles to form coherent geometric structures within a limited number of steps, often resulting in fragmented masks with high complexity. On the other hand, the 5×5 block size also degrades performance (average error $69,007.7nm^2$). While larger blocks accelerate structural changes, they are too coarse for advanced process nodes. A 5×5 flip lacks the precision required to fine-tune edge placement and often destroys necessary high-frequency details or SRAFs, making it difficult to converge to the optimal mask shape. Therefore, the 3×3 design strikes the best balance between exploration efficiency (modifying mask topology significantly) and manipulation precision (preserving fine details), confirming it as the optimal choice for our RL-ILT framework.

D. Runtime Analysis

Table IV presents the computational overhead under different training and inference configurations. It is observed that the total training time varies significantly depending on the

TABLE IV
RUNTIME AND PERFORMANCE COMPARISON UNDER DIFFERENT SETTINGS

Configuration	Total Time (h)	Inference Time (s)	$L_2(nm^2)$
w/o F	86.7	210.10	59110
w/ F w/o D	91.0	1723.44	35389
w/ F&D	127.7	14138.06	32924

strategy, ranging from 86.7 hours for the baseline pre-training to 127.7 hours for the complete pipeline incorporating data-augmented fine-tuning.

The primary source of the extended runtime in RL-ILT stems from its iterative optimization nature. Unlike supervised learning methods (e.g., Neural-ILT [7], PGAN-OPC [5]) that operate in a feed-forward manner, RL-ILT formulates mask synthesis as a sequential decision-making Markov process. The agent optimizes the mask by performing a series of discrete actions over multiple timesteps. Besides, At each timestep, the environment must invoke a lithography simulator to calculate the aerial image and provide reward feedback based on the L_2 error. This closed-loop interaction is inherently more computationally expensive than single-pass neural network inference.

While the inference time of the "Pre-training + Data-Augmented Fine-tuning" setup (approximately 3.9 hours per layout) is considerably longer than the latency of traditional deep learning models, this overhead is acceptable within the context of Mask Optimization. Unlike real-time applications, mask optimization is a backend process in semiconductor manufacturing where turnaround time is measured in hours or days, not seconds. In this domain, the priority is shifted from latency to pattern fidelity. The extended inference time is a justified trade-off for the significant improvement in L_2 error metrics and the ability to generate high-quality masks without relying on scarce ground-truth data. Moreover, the fine-tuning and inference processes for different layout clips are independent, offering significant potential for parallelization across GPU clusters, which can effectively amortize the wall-clock time in a production environment.

In summary, although RL-ILT incurs higher computational costs due to its iterative RL mechanism, the framework operates within an acceptable time domain for computational lithography, where the priority is the generation of high-fidelity mask patterns.

V. CONCLUSIONS

In this paper, we presented RL-ILT, a pixel-level reinforcement learning framework for inverse lithography that directly optimizes pattern fidelity while generating high-quality masks. By formulating mask synthesis as a Markov decision process, our approach enables end-to-end optimization through interaction with a physically accurate lithography simulator. The integration of Double DQN ensures stable and efficient learning, while action masking and 3×3 block-wise updates balance exploration granularity with mask complexity. Moreover, our proposed two-stage strategy—combining large-scale pre-training with on-the-fly data-augmented fine-

tuning—significantly enhances adaptability to diverse and previously unseen layout patterns. Experimental results validate that RL-ILT achieves superior pattern fidelity compared to existing learning-based and traditional ILT methods, as measured by squared L_2 error. This demonstrates the potential of deep reinforcement learning to address the core challenges in computational lithography of handling high-dimensional discrete design spaces.

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